DHCP Lab

Hasib Ziai

```
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ed3f:75ba:8075:5e70%10
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

thernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

thernet adapter Ethernet 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

:\WINDOWS\system32>ipconfig /renew

indows IP Configuration

o operation can be performed on Ethernet 2 while it has its media disconnected.
o operation can be performed on Ethernet 3 while it has its media disconnected.

thernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2600:8802:1100:2fe6:dcfd:7f05:405a:998b
   Temporary IPv6 Address. . . . . . : 2600:8802:1100:2fe6:8d2d:8414:9507:b2c6
   Temporary IPv6 Address. . . . . . : 2600:8802:1100:2fe6:9d65:14b5:fc93:cbb
   Link-local IPv6 Address . . . . . : fe80::dcfd:7f05:405a:998b%11
   IPv4 Address. . . . . . . . . . . : 192.168.1.116
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::b675:eff:fe35:f20a%11
                                       192.168.1.1

thernet adapter VirtualBox Host-Only Network:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ed3f:75ba:8075:5e70%10
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

thernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

thernet adapter Ethernet 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

:\WINDOWS\system32>
```

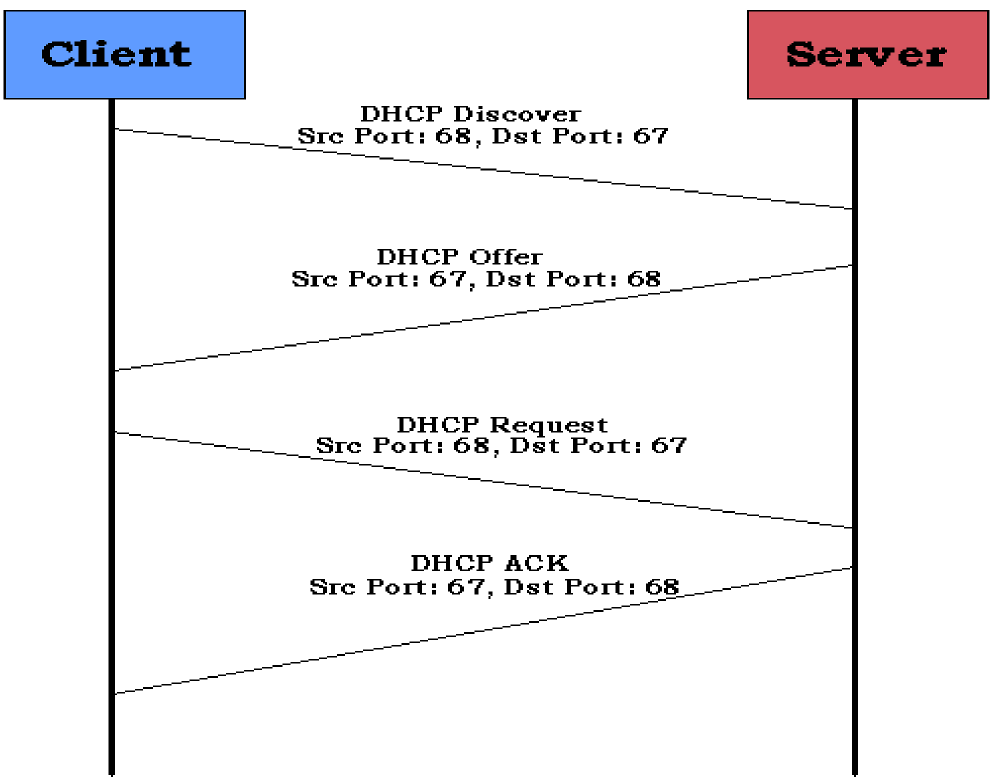
1. Are DHCP messages sent over UDP or TCP?

DHCP messages are sent via UDP as can be seen by the 4th drop-down menu, as User Datagram Protocol

```
User Datagram Protocol, Src Port: 68, Dst Port: 67
    Source Port: 68
    Destination Port: 67
    Length: 308
    Checksum: 0x2ece [unverified]
    [Checksum Status: Unverified]
    [Stream index: 27]
  > [Timestamps]
```

2. Draw a timing datagram illustrating the sequence of the first four-packet Discover/Offer/Request/ACK DHCP exchange between the client and server. For each packet, indicated the source and destination port numbers. Are the port numbers the same as in the example given in this lab assignment?

Yes, the port numbers are the same as the example given.



3. What is the link-layer (e.g., Ethernet) address of your host?

The link layer address of my host is f8:32:e4:bc:5b:9c

```
Ethernet II, Src: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  > Source: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
    Type: IPv4 (0x0800)
```

4. What values in the DHCP discover message differentiate this message from the DHCP request message?

In the DHCP request message, Option (53) is present whereas this option is not present in the DHCP discover message.

```
v Dynamic Host Configuration Protocol (Discover)
    Message type: Boot Request (1)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x444079f5
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 0.0.0.0
    Next server IP address: 0.0.0.0
    Relay agent IP address: 0.0.0.0
    Client MAC address: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
    Client hardware address padding: 00000000000000000000
    Server host name not given
    Boot file name not given
    Magic cookie: DHCP
  v Option: (53) DHCP Message Type (Discover)
      Length: 1
      DHCP: Discover (1)
  v Option: (61) Client identifier
      Length: 7

0110  00 00 00 00 00 00 63 82  53 63 35 01 01 3d 07 01   ······c· Sc5··=··
```

```
Dynamic Host Configuration Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x444079f5
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 192.168.1.116
    Next server IP address: 192.168.1.1
    Relay agent IP address: 0.0.0.0
    Client MAC address: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
    Client hardware address padding: 00000000000000000000
    Server host name: ecosystem.home.cisco.com
    Boot file name not given
    Magic cookie: DHCP
  v Option: (53) DHCP Message Type (Offer)
      Length: 1
      DHCP: Offer (2)
  v Option: (54) DHCP Server Identifier (192.168.1.1)
      Length: 4
0110  00 00 00 00 00 00 63 82  53 63 35 01 02 36 04 c0   ······c· Sc5··6··
```

5. What is the value of the Transaction-ID in each of the first four (Discover/Offer/Request/ACK) DHCP messages? What are the values of the Transaction-ID in the second set (Request/ACK) set of DHCP messages? What is the purpose of the Transaction-ID field?

The Transaction ID for the first four messages is 0x444079f5

The transaction ID for the second set of messages is 0xdbac75d1

```
Transaction ID 0x444079f5
Transaction ID 0x444079f5
Transaction ID 0x444079f5
Transaction ID 0x444079f5
Transaction ID 0xdbac75d1
Transaction ID 0xdbac75d1
Transaction ID 0xe379fa66
Transaction ID 0xa3d8ee9d
Transaction ID 0xa3d8ee9d
Transaction ID 0xa3d8ee9d
Transaction ID 0xa3d8ee9d
```

The transaction ID is to ensure that each set of requests can be differentiated from each other.

6. A host uses DHCP to obtain an IP address, among other things. But a host's IP address is not confirmed until the end of the four-message exchange! If the IP address is not set until the end of the four-message exchange, then what values are used in the IP datagrams in the four-message exchange? For each of the four DHCP messages (Discover/Offer/Request/ACK DHCP), indicate the source and destination IP addresses that are carried in the encapsulating IP datagram.

Discover: Source IP: 0.0.0.0, Destination IP: 255.255.255.255

Offer: Source: 192.168.1.1 Destination IP: 192.168.1.116

Request: Source IP: 0.0.0.0 Destination IP: 255.255.255.255
Ack: Source: 192.168.1.1 Destination IP: 192.168.1.116

7. What is the IP address of your DHCP server?
192.168.1.1

8. What IP address is the DHCP server offering to your host in the DHCP Offer message? Indicate which DHCP message contains the offered DHCP address.
The DHCP server is offering my host the IP address: 192.168.1.116

```
Dynamic Host Configuration Protocol (Offer)
    Message type: Boot Reply (2)
    Hardware type: Ethernet (0x01)
    Hardware address length: 6
    Hops: 0
    Transaction ID: 0x444079f5
    Seconds elapsed: 0
  > Bootp flags: 0x0000 (Unicast)
    Client IP address: 0.0.0.0
    Your (client) IP address: 192.168.1.116
    Next server IP address: 192.168.1.1
    Relay agent IP address: 0.0.0.0
    Client MAC address: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
```

9. In the example screenshot in this assignment, there is no relay agent between the host and the DHCP server. What values in the trace indicate the absence of a relay agent? Is there a relay agent in your experiment? If so what is the IP address of the agent?
In the example screenshot, the Relay agent IP address is 0.0.0.0, indicating no relay agent.

```
⊞ Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0 (0.0.0.0)
  Your (client) IP address: 0.0.0.0 (0.0.0.0)
  Next server IP address: 0.0.0.0 (0.0.0.0)
  Relay agent IP address: 0.0.0.0 (0.0.0.0)
  Client MAC address: Netgear_61:8e:6d (00:09:5b:61:8e:6d)
  Server host name not given
  Boot file name not given
```

There is no relay agent in my experiment either, as it is also 0.0.0.0

```
> Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 0.0.0.0
  Next server IP address: 0.0.0.0
  Relay agent IP address: 0.0.0.0
  Client MAC address: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
  Client hardware address padding: 00000000000000000000
  Server host name not given
  Boot file name not given
  Magic cookie: DHCP
```

10. Explain the purpose of the router and subnet mask lines in the DHCP offer message.
The router line is to show what the default internet gateway is on the network.
The subnet mask lines is there if a subnet within the network. The first 24 bits are the network address, and the last 8 bits is for the host address.

The first 24 bits (the number of ones in the subnet mask) are identified as the network address, with the last 8 bits (the number of remaining zeros in the subnet mask) identified as the host address. This gives you the following:

```
11000000.10101000.01111011.00000000 -- Network address (192.168.123.0)
00000000.00000000.00000000.10000100 -- Host address (000.000.000.132)
```

(Picture source:
https://support.microsoft.com/en-us/help/164015/understanding-tcp-ip-addressing-and-subnetting-basics)

11. In the DHCP trace file noted in footnote 2, the DHCP server offers a specific IP address to the client (see also question 8. above). In the client's response to the first server OFFER message, does the client accept this IP address? Where in the client's RESPONSE is the client's requested address?
Yes, the client accepts this IP address. This can be noted in the DHCP Request, in which the client sends a request for the offered IP address. (Option 50)

```
Option: (53) DHCP Message Type (Request)
    Length: 1
    DHCP: Request (3)
Option: (61) Client identifier
    Length: 7
    Hardware type: Ethernet (0x01)
    Client MAC address: ASUSTekC_bc:5b:9c (f8:32:e4:bc:5b:9c)
Option: (50) Requested IP Address (192.168.1.116)
    Length: 4
    Requested IP Address: 192.168.1.116
Option: (54) DHCP Server Identifier (192.168.1.1)
    Length: 4
    DHCP Server Identifier: 192.168.1.1
Option: (12) Host Name
    Length: 12
    Host Name: HasibDesktop
```

12. Explain the purpose of the lease time. How long is the lease time in your experiment?
The lease time is how long the client can use the given IP address. The lease time in my experiment is 86400 seconds, or 1 day.

```
Option: (54) DHCP Server Identifier (192.168.1.1)
    Length: 4
    DHCP Server Identifier: 192.168.1.1
Option: (51) IP Address Lease Time
    Length: 4
    IP Address Lease Time: (86400s) 1 day
Option: (58) Renewal Time Value
    Length: 4
    Renewal Time Value: (43200s) 12 hours
Option: (59) Rebinding Time Value
    Length: 4
```

13. What is the purpose of the DHCP release message? Does the DHCP server issue an acknowledgment of receipt of the client's DHCP request? What would happen if the client's DHCP release message is lost?
The DHCP release message is used by the server to release the assigned IP address.
The DHCP does send an ACK after receiving the client's DHCP release request.

807	21.015239	192.168.1.116	192.168.1.1	DHCP	352	DHCP Request - Transaction ID 0xdbac75d1
809	21.019587	192.168.1.1	192.168.1.116	DHCP	359	DHCP ACK - Transaction ID 0xdbac75d1
885	33.151990	192.168.1.116	192.168.1.1	DHCP	342	DHCP Release - Transaction ID 0xe379fa66

If the client's DHCP release message is lost, the server would not know this and the client would release the IP address but the server would not release the client's stored IP address until it's lease expires.

14. Clear the bootp filter from your Wireshark window. Were any ARP packets sent or received during the DHCP packet-exchange period? If so, explain the purpose of those ARP packets.

Yes, there were ARP packets both sent and received during the DHCP packet-exchange period. It seems like the client (my computer) sent an ARP request onto the network so that it could communicate with other devices on the network, via a physical MAC address for the IP.

191	8.678020	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.116
192	8.678222	BelkinIn_35:f2:0a	ASUSTekC_bc:5b:9c	ARP	60	192.168.1.1 is at b4:75:0e:35:f2:0a
193	8.678782	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.116
194	8.678930	BelkinIn_35:f2:0a	ASUSTekC_bc:5b:9c	ARP	60	192.168.1.1 is at b4:75:0e:35:f2:0a
234	9.046595	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
276	10.045926	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
371	11.046249	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
470	12.046578	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	ARP Announcement for 192.168.1.116
1022	39.004290	BelkinIn_35:f2:0a	Broadcast	ARP	60	Who has 192.168.1.143? Tell 192.168.1.1
1034	39.763864	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.116
1035	39.764176	BelkinIn_35:f2:0a	ASUSTekC_bc:5b:9c	ARP	60	192.168.1.1 is at b4:75:0e:35:f2:0a
1048	39.778060	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.116
1049	39.778481	BelkinIn_35:f2:0a	ASUSTekC_bc:5b:9c	ARP	60	192.168.1.1 is at b4:75:0e:35:f2:0a
1063	40.045722	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
1108	41.046048	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
1166	42.046376	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	Who has 192.168.1.116? (ARP Probe)
1253	43.045712	ASUSTekC_bc:5b:9c	Broadcast	ARP	42	ARP Announcement for 192.168.1.116